

PHYSICS

UNIT I – Lasers, Fiber Optics & Ultrasonics

Lasers

- Characteristics of lasers
- Stimulated emission
- Population inversion
- Einstein's coefficients
- CO₂ laser
- Semiconductor laser
- Laser-Induced Breakdown Spectroscopy (LIBS)
- Instrumentation
- Engineering applications of lasers

Fiber Optics

- Construction of optical fiber
- Types of optical fibers
- Refractive index profiles
- Fiber drawing process
- Double cladding method
- Basic principles of fiber optic sensors
- Construction and working of:

- Pressure sensors
- Applications of optical fibers in engineering

Ultrasonics

- Introduction to ultrasonic waves
- Production of ultrasonic waves:
 - Magnetostriction method
- Ultrasonic pulse-echo testing method
- Engineering applications of ultrasonics

UNIT II – Semiconductor Physics & EM Theory

Semiconductor Physics

- Types of semiconductors
- Direct and indirect bandgap semiconductors
- Hall effect
- Construction and working of:
 - Quantum well
 - Light Emitting Diodes (LEDs)
 - Solar cells
- Applications of semiconductor devices
- Energy harvesting devices:
 - Piezoelectric generators

- Thermoelectric generators
- Properties and advantages of graphene

Electromagnetic Theory

- Basic laws of electricity and magnetism
 - Displacement current
 - Maxwell's equations
 - Integral and differential forms of Maxwell's equations
 - Poynting theorem
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UNIT III – Magnetic Materials & Superconductors

Magnetic Materials

- Types of magnetic materials
- Weiss molecular field theory
- Magnetic domains
- Hysteresis curve
- Soft and hard magnetic materials
- Magnetoresistance materials (CMR & GMR)
- Applications of magnetic materials

Superconductors

- Properties of superconductors
- Meissner effect

- Type I and Type II superconductors
 - BCS theory (qualitative)
 - High-T_c superconductors
 - Applications of superconductors
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UNIT IV – Wave Mechanics & Quantum Computation

Wave Mechanics

- Matter waves
- de Broglie wavelength
- Physical significance of wave function
- Schrödinger time-independent wave equation
- Energy of a particle in:
 - One-dimensional potential box
- Kronig–Penney model (qualitative)

Quantum Computation

- Introduction to quantum computing
- Classical bits and qubits
- Basics of quantum gates:
 - Hadamard
 - CNOT
- No-cloning theorem

- Quantum teleportation (basic idea)
- Applications of quantum computing

- Applications of thin films

UNIT V – Nano Materials & Thin Films

Nanomaterials

- Introduction and properties of materials at reduced size
- Surface-to-volume ratio at nanoscale
- Classification of nanomaterials
- Preparation techniques:
 - Bottom-up method (Sol-Gel)
 - Top-down method (Ball milling)
- Characterization techniques:
 - X-ray diffraction (XRD)
 - Scanning Electron Microscope (SEM)
 - Transmission Electron Microscope (TEM)
- Applications of nanomaterials

Thin Films

- Distinction between bulk and thin films
- Thin film preparation techniques:
 - Thermal evaporation method
 - Electron beam evaporation method